CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 5: Backtracking & Branch-and-Bound Paradigms

Course: Design Analysis and Algorithms [23CS3502]
Category: Assignments | Assignment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Systematic state space tree exploration with bounding function pruning.

2. Assignment Problems & Instructions:

1. Draw the complete state space tree for 4-Queens problem using backtracking.
2. Formulate the bounding function for m-Coloring graph problem.
3. Solve the 15-Puzzle problem using Least Cost Branch and Bound LCBB .
4. Compare computational complexity of Backtracking vs Branch-and-Bound for 0/1 Knapsack.

Submission Guidelines & Verification:

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